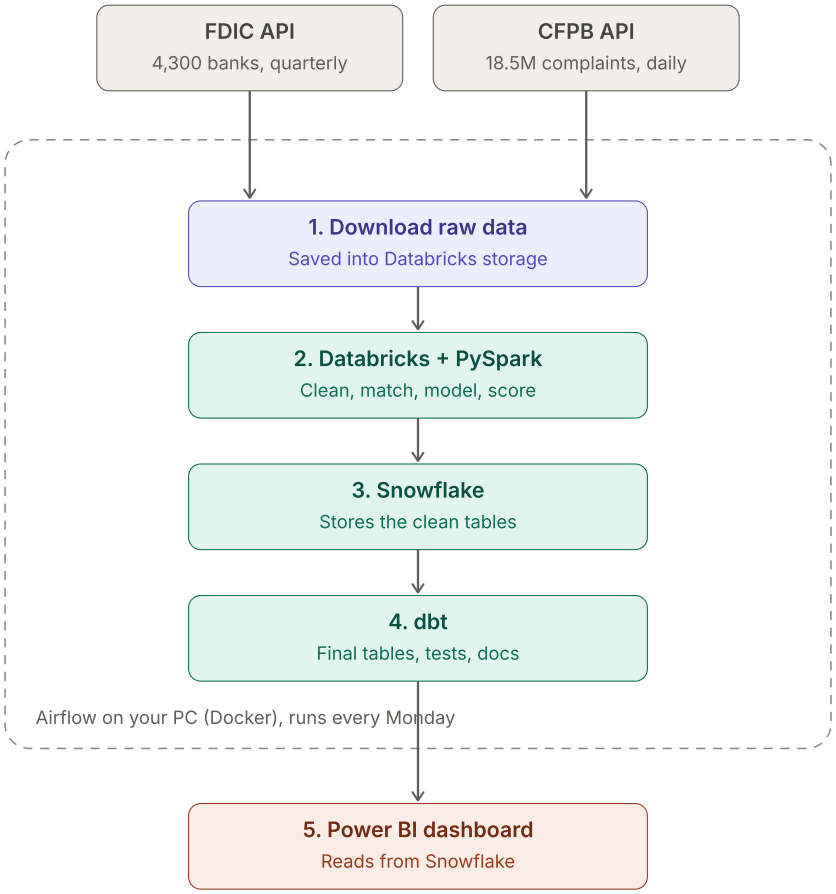
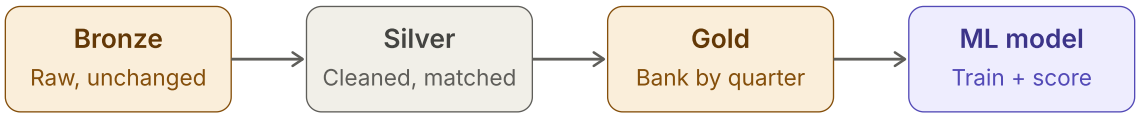
Bank Health Data Platform: workflow

FDIC bank financials + CFPB consumer complaints → Airflow, Databricks (PySpark, MLflow), Snowflake, dbt, Power BI · Martins Ogiekpolor

End-to-end pipeline



Inside step 2: Databricks



What one weekly run does

1. **Download.** Airflow fetches the newest bank financials from the FDIC and new complaints from the CFPB, and drops the files into Databricks.
2. **Databricks and PySpark.** The files are stored as-is (bronze), cleaned and matched to the right bank (silver), and summed up per bank per quarter (gold). The model then gives every bank a 12-month risk score, and MLflow records each training run.
3. **Snowflake.** The clean tables and risk scores are copied into the Snowflake warehouse.
4. **dbt.** Builds the final reporting tables inside Snowflake and runs tests, so wrong numbers stop the run before they reach the dashboard.
5. **Power BI.** Reads those final tables to show the dashboard.

Steps 1 to 4 run by themselves every Monday while Docker is running on the PC. Step 5 refreshes when Power BI is opened and Refresh is clicked.